



Document purpose: External-facing FAQ to support demonstrations, partner engagements, and post-demo follow-up.

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1. General & Technical

1. What is SPICE?

SPICE is an offline-first digital health application and workflow platform designed to support community-level primary health care, including screening, registration, longitudinal follow-up, clinical decision support, and program analytics.

2. Is SPICE compatible with other healthcare systems and platforms?

Yes. SPICE supports standard interoperability protocols (FHIR and HL7). It has the capability to exchange with FHIR-ready systems in any digital health environment

3. Can workflows in SPICE be customized to suit my organization?

Yes. Non-clinical and administrative workflows are customizable. Clinical decision-support algorithms are based on established guidelines and are not customizable; however, parameters and non-clinical process flows can be adapted to local needs.

4. Is SPICE open source?

SPICE is available as open-source packages with options for hosted (managed) or self-managed deployments. Release notifications are distributed through official channels for open-source users. The codebase is accessible on the github link: <https://github.com/Medtronic-LABS/spice-2.0-android>. Deployment instructions are also available through gitbook on the link: <https://spice.docs.medtroniclabs.org/deploy/deployment-guide>

5. Can SPICE work offline?

Yes, partially. SPICE is built with an offline-first architecture, but offline functionality is workflow-specific.

All health worker profiles can use the Screening workflow offline at the community or facility level. Health workers can capture screening data without internet connectivity and synchronize the data once an internet connection becomes available.

The other workflows require connectivity to ensure data integrity, synchronization, and coordination across users and facilities.

6. How does SPICE support clinical decision making?

SPICE supports clinical decision-making through embedded, evidence-based decision-support logic that operates within routine care workflows. The platform integrates WHO-recommended algorithms (including WHO HEARTS) and can be configured to align with approved national and local clinical guidelines where these are formally adopted by the implementing health authority.

As health workers enter patient data, SPICE automatically calculates key clinical indicators such as BMI, blood pressure categories, cardiovascular (CVD) risk scores, and other condition-specific measures. Based on these inputs, the system provides context-specific prompts and structured guidance to support appropriate actions, including referrals, treatment planning, follow-up scheduling, and escalation of care.

Clinical decision support in SPICE is workflow-driven and on-demand. Guidance is surfaced at the point in the workflow where a decision is required, such as during screening, assessment, enrollment, or follow-up, rather than as a standalone tool. This ensures that decision support is available in real time, within the care process, without disrupting service delivery.

Beyond hypertension and type 2 diabetes, SPICE supports additional service areas, including mental health, maternal health, and other primary health care domains, through structured workflows and decision logic aligned to the scope of each implementation. Where applicable, decision support is configured to reflect the care pathways and data requirements defined by the health system. Support for conditions such as Type 1 diabetes is guided by national clinical protocols and implementation scope, rather than generic automation.

Importantly, SPICE does not replace clinical judgment. Instead, it standardizes routine decision points, reduces variation, and reinforces adherence to approved guidelines, enabling health workers to make consistent, evidence-informed decisions while retaining full clinical responsibility.

7. How does SPICE enhance data quality?

SPICE improves data quality through structured digital workflows, mandatory fields, built-in validation checks, and role-based access controls. Real-time prompts reduce incomplete or inconsistent entries, while audit logs and dashboards support routine review, supervision, and corrective action. SPICE also employs menus options for data entry on common content, and users only need to type in what is not captured on the menus.

8. How does SPICE support longitudinal patient management?

SPICE maintains a single, longitudinal patient record. Once a patient is enrolled, the platform schedules reviews, tracks follow-ups and referrals, flags overdue visits, and records outcomes, enabling continuity of care across community outreach and facility encounters.

9. Can SPICE export data for reporting or analysis?

Yes. SPICE supports data export and reporting in formats agreed with the implementing authority. Data exports are governed by role-based permissions and data protection requirements and are typically used for routine reporting, supervision, monitoring, and approved analyses.

2. Data Protection, Privacy & Security

10. Who owns the data collected in SPICE?

Data ownership rests with the implementing health authority or facility (the data controller). Medtronic Labs operates as a software provider/data processor under agreed governance arrangements and does not claim ownership of program data.

11. What is the compliance state of Medtronic Labs to local Data Privacy regulations?

Medtronic LABS is compliant with applicable local data protection and privacy regulations in countries of operation. The organization operates as a data processor on behalf of the implementing health authority and applies standardized data protection and security controls aligned with national laws and recognized international standards.

11. Does SPICE store data outside the country of implementation?

Hosting and data-residency options are configured per implementation agreement. SPICE can be hosted locally or in approved regional/cloud environments to meet national data residency and regulatory requirements.

12. What personal data does SPICE collect?

SPICE collects the minimum personal data required for care delivery: identifiers (name or program ID), demographic information, clinical measurements, and program-linked IDs. Implementations follow data-minimization principles.

13. What is considered sensitive personal data in SPICE?

Health-related information, including diagnoses, treatment notes, and bio measures (e.g. blood pressure, blood glucose, height, weight), is sensitive personal data and receives enhanced protection.

14. How does SPICE protect Personal Health Information (PHI)?

SPICE safeguards PHI with strong encryption in transit and at rest, role-based access control and authentication, audit logging, secure APIs (TLS/HTTPS), regular security testing (including vulnerability assessments), and a documented incident response process.

15. What happens to data when it is stored (data at rest) in SPICE?

Data at rest is encrypted and protected by strict access controls. Stored patient information is unreadable to unauthorized users. All access and changes are logged for accountability. Handling of stored data complies with applicable data-protection laws and the hosting/governance arrangements agreed with the implementing authority.

16. How are consent and privacy notices handled?

Before SPICE is used for care delivery, a consent is obtained in accordance with local legal and program requirements. Privacy notices are presented at assessment/enrollment points and consent is captured via opt-in mechanisms configurable to local regulations and program protocols.

17. How does SPICE handle subject access, rectification, and erasure requests?

These requests are managed by the data controller. SPICE provides technical features (data export, correction of workflows, deletion and retention controls) to support controllers in fulfilling legal obligations.

18. What happens if a device used for SPICE is lost or stolen?

App data on devices is encrypted, and access is protected by authentication. Credentials can be revoked centrally to prevent further synchronization. Implementers should follow device-management and incident-reporting procedures to revoke access and, where appropriate, wipe device data.

19. How long is patient data retained in SPICE?

Retention periods are defined by the data controller and comply with national laws and program policies. SPICE supports configurable retention and archival settings.

20. What audit and logging features do SPICE provide?

SPICE maintains detailed audit logs that record user access, data changes, and synchronization events to support monitoring, investigations, and accountability.

21. How are incidents and data breaches handled?

SPICE follows a documented incident-response process: contain, investigate, notify the data controller and relevant authorities as required by law, remediate, and report. Partners can request the incident-response SOP for details.

22. Can SPICE data be used for research or evaluations?

Yes, but only with appropriate approvals. Use of SPICE data for research or secondary analysis requires authorization from the data controller and must comply with national data protection laws, ethical review requirements, and institutional approvals. Where required, data can be anonymized or de-identified before use.

23. What approvals are required for secondary use of SPICE data?

Secondary use of SPICE data (e.g., for research, learning, or publication) typically requires approval from the implementing authority, ethics committees (where applicable), and adherence to national data protection regulations. SPICE provides technical controls to support approved data access and extraction.

3. Frontline Health Workers

24. Who should use SPICE?

SPICE is intended for Community Health Nurses/Officers (CHNs/CHOs), facility nurses, clinicians, health information/records officers, laboratory staff, pharmacy staff, nutrition/lifestyle officers, and supervisors. Each user sees only the workflows relevant to their assigned role.

25. Does SPICE replace Electronic Medical Records (EMRs) or paper records?

No. SPICE complements existing EMRs and paper systems by strengthening screening, referral, longitudinal follow-up, and community-to-facility linkage. Managers decide how SPICE fits alongside existing records and reporting requirements.

26. I forgot my password. What should I do?

Use the “Forgot Password” option on the login screen. If that fails, contact your facility SPICE focal person or district ICT support for account recovery.

27. Can I share my login details with a colleague?

No. Each SPICE account is personal and traces data entry to the individual user. Sharing credentials is not permitted and may breach data-protection and accountability rules.

28. Why can't I see some workflows in my account?

Workflow access is role-based. If you cannot see a workflow, it may not be assigned to your role, or your account setup may be incomplete. Contact your supervisor or SPICE support.

29. Why is client consent required?

Consent protects patients' rights, meets legal and ethical requirements, builds trust, and ensures patients understand how their information will be used. Consent must be obtained before enrolling a patient into SPICE.

30. What if a client refuses consent?

Respect their decision. Do not enroll them in SPICE. Continue to provide routine care and document care per facility procedures.

31. Can I sign consent on behalf of a client?

Only if the patient is unable to sign and facility protocol permits verbal proxy consent with appropriate witness documentation. Always follow facility guidance and legal requirements.

32. Who should be screened?

Routine screening targets adults (typically 30 years and older), people with symptoms suggestive of diabetes or hypertension, and those with risk factors (obesity, family history, tobacco use, sedentary lifestyle).

33. How many blood pressure (BP) readings should I take?

Follow SPICE prompts and local clinical protocols. Generally, two to three properly performed readings are recommended. Ensure correct patient positioning and device technique.

34. What happens after screening?

SPICE provides a screening outcome (normal, refer for assessment, or urgent referral). Explain the result to the patient and follow recommended next steps (monitor, refer, or urgent care).

35. What is the difference between screening and enrollment?

Screening identifies potential risk or abnormal findings. Enrollment (registration) creates a permanent patient record for longitudinal follow-up and management once follow-up or diagnosis is required.

36. What if I cannot find a patient in SPICE?

Verify spellings, search by phone number or program ID, and check alternate identifiers. If the patient truly does not exist, register them per the enrollment workflow. Avoid creating duplicate records.

37. When should I conduct an assessment?

Conduct assessments during scheduled follow-up visits, community outreach encounters, when patients report new symptoms, or when SPICE flags a patient as overdue or a defaulter.

38. Who is a defaulter?

A defaulter is a patient who missed a scheduled follow-up appointment. SPICE flags defaulters so outreach and tracing can be prioritized.

39. What does LTFU in SPICE mean?

LTFU stands for Lost to Follow-Up, patients who have not returned for care within an expected timeframe, usually 90 days, and may require active tracing.

40. What should I do if a BP or BG reading is very high?

Do not ignore alerts. Reconfirm measurements, explain the risk to the patient, provide immediate advice per protocol, and refer or escalate care as required. Notify your supervisor when appropriate.

41. Who can prescribe medication on SPICE?

Only authorized clinicians with prescribing privileges can enter prescriptions or treatment orders. Pharmacy dispensing must reflect authorized prescriptions.

42. What if the pharmacy cannot dispense the full prescription?

Dispense what is available, record the correct reason (e.g., stock-out), and inform the clinical team. Do not enter false values.

43. Can lab results be edited after submission?

Lab results should not be altered casually. Corrections must follow the approved correction workflow and be recorded with a reason for audit-trail integrity.

44. What should I do if I make a data entry mistake?

Correct the error immediately using the system's correction workflow if available. If the mistake affects clinical care, inform your supervisor and document the correction as required.

45. Why is a patient's correct phone number important?

Accurate phone numbers enable follow-up calls, appointment reminders, and outreach tracing. Incorrect numbers increase loss to follow-up and reduce program effectiveness.

46. What logistics are required for full operationalization of SPICE by health workers?

Minimum practical requirements: an Android smartphone or tablet that runs SPICE; charger and power bank where electricity is unreliable; essential clinical tools (BP apparatus, glucometer, scale, height measure); initial training and periodic refreshers; supervisory oversight. Internet is not required for routine use but is needed periodically for syncing, updates, and remote supervision.

47. Do health workers have to re-register patients at every encounter?

No. Patients are registered once when enrolled. On subsequent visits, health workers search for and select the existing patient record to conduct follow-up, assessment, or review.

48. Can a patient who is registered at one facility be retrieved and managed at another facility?

Yes. Where allowed by the implementing health system's configuration and data-governance policies. Cross-facility retrieval supports referrals, continuity of care, and patient mobility without re-registration.

49. How does SPICE improve continuity of care at the community level?

SPICE enables Community Health Nurses (CHNs) to track individuals from screening through referral and follow-up; it generates task lists for patients due for review, flags missed visits and defaulters and maintains a single patient record that follows the patient across outreach and facility encounters, reducing loss to follow-up and strengthening linkages.

50. What happens if a trained health worker is transferred or leaves the facility?

When a health worker is transferred or exits service, their SPICE account would be deactivated in line with facility procedures. Patient records remain within the system and are accessible, ensuring continuity of care and preventing data loss.

51. How are new users onboarded onto SPICE?

New users are onboarded through role-based account creation and training. Access is granted according to the user's role (e.g., CHN, clinician, pharmacy, supervisor), ensuring users only see workflows relevant to their responsibilities.

52. Can SPICE support patient education or counseling documentation?

Yes. SPICE allows health workers to document counseling, lifestyle advice, and patient education as part of routine care records. This supports continuity of care and reinforces adherence and behavior change over time.

53. Who do I contact for technical support?

Follow the local escalation path:

L1 — Facility SPICE focal person

L2 — District ICT support

L3 — Regional support

L4 — National SPICE support / SPICE team

General enquiries: community@medtroniclabs.org

4. Implementation, Evidence & Scale-up (Ghana-specific)

54: Has SPICE been tested in real healthcare delivery conditions?

Yes. SPICE has been implemented across 134 GHS and CHAG facilities in Ghana (2021–2026), embedded within routine services and independently evaluated in a mixed-methods process evaluation across eight regions.

55. Can frontline health workers use SPICE as part of routine work?

Yes. An independent evaluation found SPICE to be intuitive and easy to use. Over 2,000 health workers have been trained, and users reported that minimal supervision is required after initial training.

56. Does SPICE improve continuity of care and follow-up?

Yes. An independent evaluation showed improved patient follow-up and adherence. Over 90,000 enrolled patients had documented follow-up data; evaluated cohorts showed improvements in hypertension and diabetes metrics associated with structured digital follow-up.

57. Is SPICE aligned with national primary health care models?

Yes. Deployments are embedded within GHS and CHAG structures and aligned with community screening, referral, enrollment, and longitudinal care models. Further, SPICE application follows the country defined health domain protocols and in cases where those do not exist, references established protocols from international authorities such as WHO.

58. What operational challenges were identified during PoC deployments?

Common challenges include medication unavailability, handling existing staff workload, frequent staff rotation, and parallel documentation. Mitigations include refresher training, mentorship, and supervision. These are system-level issues to address during scale-up planning.

5. Demo & Operational Questions

59. What happens after a demo if participants request the app or APK?

The SPICE APK and demo materials may be shared subject to implementing partner distribution policies and data-controller approval. For hosted demos, the SPICE team provisions demo instances as agreed.

60. Can SPICE be deployed quickly?

Deployment speed depends on scope. Basic deployments can begin within weeks using standard Android devices and focused training. Full roll-out is typically phased with training, supervision, and governance alignment.

61. Who can access dashboards and management reports?

Dashboard access is role-based. Administrators, supervisors, and managers receive views appropriate to their responsibilities via role-based permissions.

6. Partnerships, Governance & Sustainability

62. What does it mean when SPICE is described as “free and open source” for governments? Are there hidden costs?

SPICE is made available to governments as an open-source digital public good, meaning there are no licensing or usage fees for the software itself. Governments can access, deploy, and adapt the platform without paying for proprietary licenses.

However, as with any digital system, operational costs still apply. These may include hosting, devices, connectivity, training, supervision, customization aligned to national guidelines, and ongoing system support. These costs are transparent, implementation-specific, and typically budgeted as part of program or government operational planning. There are no hidden fees embedded in SPICE.

63. When we say government can “own” SPICE, what does ownership mean in practice?

Ownership means that the government controls how SPICE is deployed, governed, and used within its health system. This includes control over data, workflows, hosting decisions, and integration with national digital architecture.

Because SPICE is open source, governments may choose to host and manage the system themselves or engage trusted partners to do so. Ownership does not require Medtronic Labs to retain control over the system or data.

64. Does ownership mean Medtronic Labs transfers the SPICE source code to government?

SPICE’s source code is already openly available under an open-source license. Governments do not need a special transfer to access it.

What matters in practice is operational stewardship, not code possession alone. Medtronic Labs supports governments to deploy, configure, and govern SPICE in line with national policies, even where governments choose not to directly manage the codebase themselves.

65. What role does Medtronic Labs play after SPICE is adopted by the government?

Medtronic Labs’ role shifts from implementer to technical, programmatic, and analytics partner, depending on government needs. This may include support with configuration, analytics, outcomes measurement, capacity building, and system optimization.

Medtronic Labs does not position itself as a perpetual system owner. The long-term intent is to build government capacity and institutional ownership, with Medtronic Labs providing value-added support where requested.

66. Why does Medtronic Labs offer SPICE to governments at no licensing cost? What does Medtronic Labs gain?

Medtronic Labs' mission is focused on health system strengthening and improved outcomes, not software licensing revenue. Offering SPICE as open source reduces barriers to adoption, avoids vendor lock-in, and supports sustainability.

Medtronic Labs' value proposition lies in implementation experience, analytics, outcomes expertise, and system-level insight, rather than proprietary control of technology. Trust, long-term partnerships, and impact at scale are core drivers—not licensing income.

67. Can SPICE support co-fundraising or joint resource mobilization with governments and partners?

Yes. SPICE is frequently positioned as part of a co-funded health system strengthening package, where governments, donors, and partners jointly mobilize resources for service delivery, training, analytics, and supervision.

Because SPICE is open source, it can be embedded within donor-funded or government-funded programs without creating dependency on a single vendor, making it attractive for pooled or blended financing models.

68. Can SPICE be linked to national identification systems?

SPICE can be configured to capture national ID numbers where policy and regulatory approvals exist. Technical linkage to national ID systems depends on government authorization, interoperability agreements, and national data protection regulations.

SPICE does not automatically integrate with national ID systems without explicit government approval and governance frameworks.

69. If a patient has already been screened or registered in SPICE, will the system detect this if they present at another location?

Yes. SPICE is designed to prevent duplicate records by enabling patient search using identifiers such as name, phone number, program ID, or other approved identifiers. If a patient already exists in the system, health workers can retrieve the existing record rather than creating a new one.

This supports continuity of care across locations, subject to access permissions and governance rules.

70. If a government already has a system (e.g. e-Tracker in Ghana), why not enhance it instead of introducing SPICE?

Medtronic Labs does not position SPICE as a replacement for national systems by default. The decision to adopt, integrate, or run systems in parallel rests with government.

In practice, some national systems are optimized for reporting or program monitoring, while SPICE is designed for frontline longitudinal patient management and decision support. Where appropriate, SPICE can complement existing systems through interoperability rather than duplication.

71. Is there a real difference between SPICE and systems like e-Tracker?

Yes. While both support health data capture, they are designed for different primary purposes. SPICE focuses on patient-level longitudinal care, decision support, follow-up tracking, and community–facility linkage, whereas systems like e-Tracker are often designed for reporting and program monitoring.

Governments determine how these systems coexist—through integration, complementary use, or phased alignment—based on national digital health strategy.

72. How does SPICE align with national digital health governance and HIE initiatives?

SPICE is designed to operate within national digital health architectures, including Health Information Exchanges (HIEs). It supports interoperability standards (e.g. FHIR) and can contribute high-quality frontline data into national data backbones without creating parallel silos.

Alignment is driven by government policy, not by unilateral system deployment.

73. If SPICE is positioned as a community-level digital health system, how does it support continuity of care when patients are referred to higher-level facilities?

SPICE supports end-to-end, longitudinal care by exchanging patient data through national Health Information Exchanges (HIEs). Data captured at the community level can be made available to higher-level facilities through the HIE, enabling clinicians to view prior patient information at the point of care.

Care continuity is bi-directional. Information generated at higher-level facilities, including diagnoses, treatment decisions, and follow-up instructions, can be shared back through the HIE, allowing community-level health workers to continue follow-up and monitoring using up-to-date clinical information.

In settings where higher-level facilities do not yet have an Electronic Medical Record (EMR) system, SPICE can provide a light-touch EMR capability to support essential patient documentation, referrals, and follow-up. This approach ensures continuity of care without introducing parallel or siloed systems and allows facilities to participate in national data exchange while longer-term EMR strategies are developed.

SPICE is designed to complement existing national digital health architectures, strengthening linkage between community and facility care while supporting interoperability, system ownership, and sustainability.

7. Support and Releases

74. How often is SPICE updated, and are there costs for updates?

SPICE is updated regularly. Hosted implementations typically receive updates at no additional cost per the support agreement. Open-source deployments receive release notifications; independent implementers manage their own updates.

75. What support channels are available?

Support resources include online documentation, issues reporting through the app and helpdesk, role-based guides, GitHub discussions, and direct support via community@medtroniclabs.org. Support SLAs are defined in implementation agreements for hosted deployments.

8. Contacts

For general enquiries and support: community@medtroniclabs.org. For data protection enquiries: contact the Implementing Program's Data Protection Officer (contact details supplied per deployment)

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